

Aninternshipin
DataAnalyticswithTableau

by

SmartInternz

ProjectName: Visualizinghousingmarkettrends:ananalysisofsalepricesand features

TeamMembers:

- | | |
|------------------------------|---------------------|
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ABSTRACT

This project, titled “Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau,” focuses on transforming complex real estate data into clear, actionable insights through interactive data visualization. By cleaning and preparing a dataset containing various housing attributes—such as sale price, area, number of bedrooms, renovation status, and location—key trends were uncovered using Tableau’s powerful visual analytics. The project involved the creation of calculated fields (e.g., TotalAreaSqft, SalePriceBin), the use of filters (e.g., condition, renovation status, zipcode group), and the development of dashboards and stories that narrate insights across multiple dimensions. These dashboards were then embedded into a Flask web application, ensuring easy accessibility and deployment. The resulting solution empowers users—including buyers, real estate agents, and policy makers—to make data-driven decisions. With its scalability and modular structure, the project lays a foundation for further enhancements like live data integration, predictive analytics, and expanded geographic coverage.

Key Words:

- TableauDashboard
- HousingMarketAnalysis
- DataVisualization
- SalePricePrediction
- PropertyFeatures
- RenovationInsights

ProjectReportFormat

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ProposedSolution

Solution Architecture

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Link

GitHub&ProjectDemoLink

1. Introduction

The real estate market is influenced by various factors such as house age, renovation status, number of bedrooms and bathrooms, and overall size. This project aims to analyze housing market trends and visualize key insights using Tableau to better understand how different features impact sale prices.

Project overview

The dataset contains transformed housing data and 21,609 house sale records, including property features such as sales price, area, bedrooms, bathrooms, floors, and location. There are a total of 31 columns, out of which Sale Price can be supposedly taken as a dependent variable. The other variables are different features, locations and date, etc. regarding the houses. This project, "Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau," aims to explore and analyze housing market trends using the Transformed Housing Data 2 dataset from Kaggle. The objective is to identify key factors influencing house prices, such as location, size, number of bedrooms, bathrooms, floors, and basement area.

By leveraging Tableau, the project will create interactive dashboards, story, bar chart, histogram, summary dashboard to visualize patterns, compare regional price variations, and gain insights into how different features impact house sale prices. The analysis will help in making data-driven decisions for buyers, sellers, and real estate professionals.

Objectives

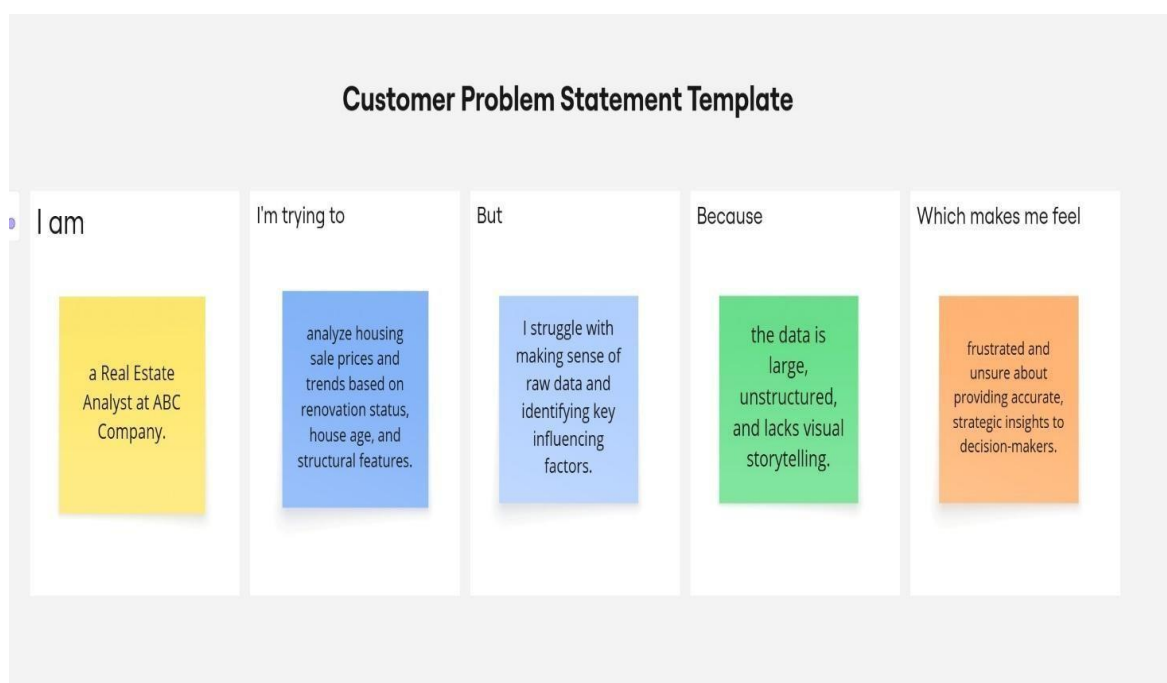
- Identify key factors influencing house prices.
- Analyze the effect of renovations on property value.
- Explore the distribution of house sales across different price ranges.
- Create interactive Tableau dashboards to present findings effectively.

2. Project Initialization and Planning Phase

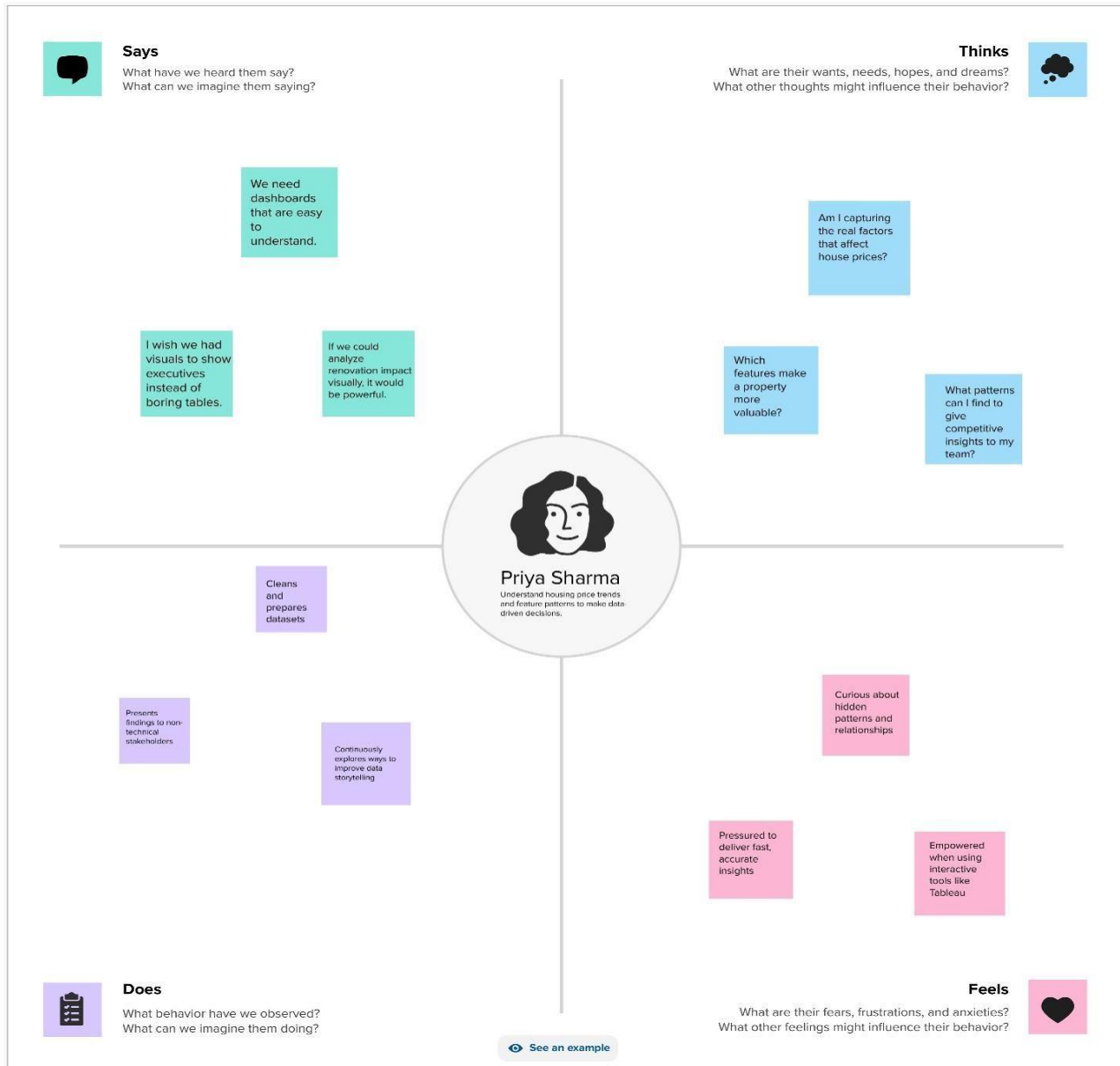
Define Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a Real Estate Analyst at ABC Company	analyze housing sale prices and trends based on renovation status, house age, and structural features.	I struggle with making sense of raw data and identifying key influencing factors.	the data is large, unstructured, and lacks visual storytelling.	frustrated and unsure about providing accurate, strategic insights to decision-makers.
PS-2	a Company Executive at ABC Company	make data-driven decisions to optimize housing pricing strategies.	I cannot easily interpret detailed data or compare features like bathrooms, floors, and renovations.	traditional reports are too complex and not visually accessible.	disconnected from the real trends in the housing market and hesitant to make confident business decisions.

Problem Statement Template:



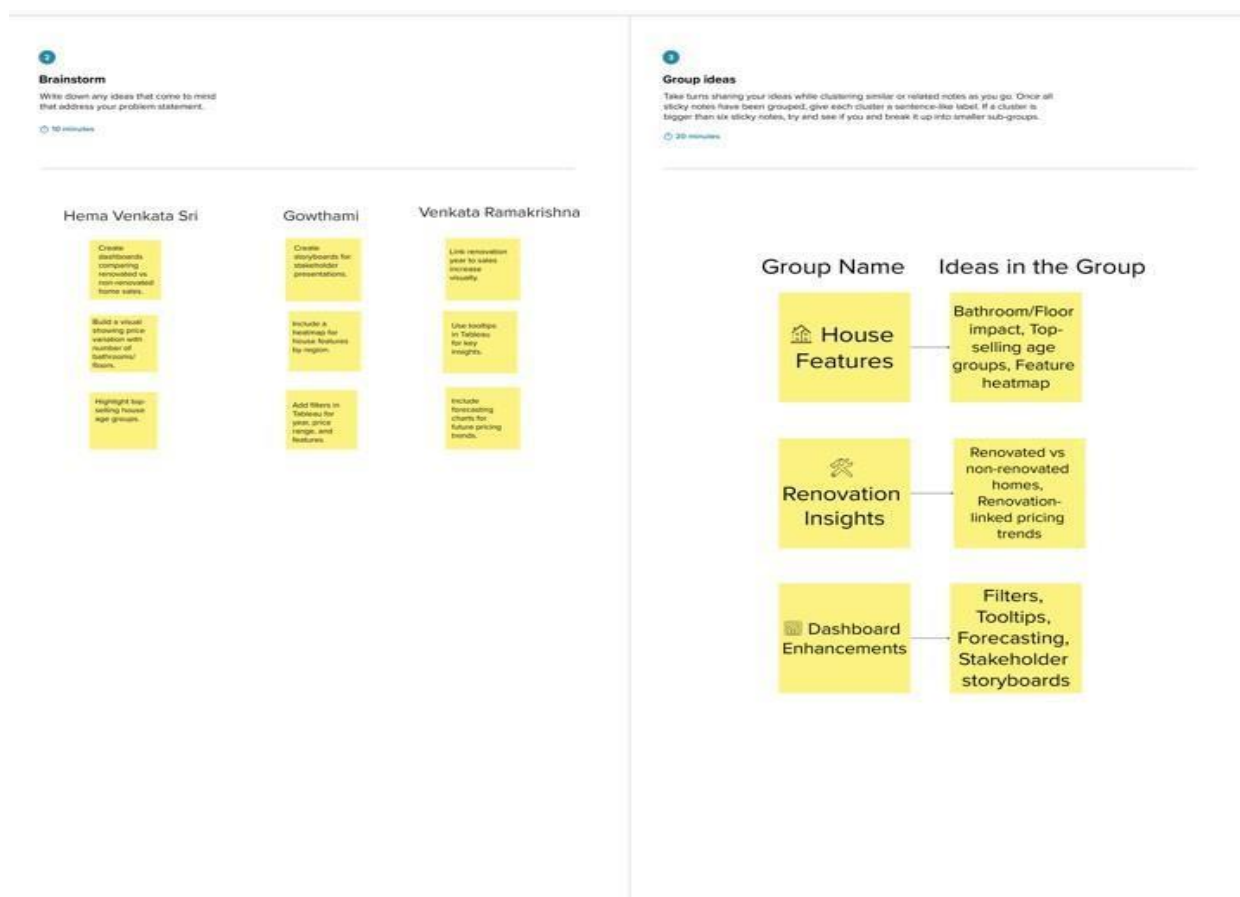
2.2 EmpathyMapCanvas



2.3 Brainstorming

Step 1: Team Gathering, Collaboration and Problem Statement

Our team collaborated to identify pressing challenges in the real estate market, particularly in understanding how various property features influence housing sale prices. After exploring themes like housing affordability, real estate investment planning, urban development, and smart property insights, we narrowed down our focus to uncover actionable insights hidden in housing data. The objective was to visually explore trends using Tableau that would help buyers, sellers, investors, and policy makers understand patterns of sale prices based on features like area, bedrooms, renovation status, condition, location (zipcode groups), and more.



ProblemStatement:

Howcanhousingsalepricetrendsandpropertycharacteristicsbevisualizedandanalyzedusing Tableau to identify patterns, improve buyer/seller decision-making, and uncover insights that support strategic real estate planning?

TeamMembers:

- TeamLeader:AnnamdevulaHemaVenkataSri Team
- Member: Dola Gowthami
- TeamMember:Chillara Venkata Ramakrishna

Step 2:Brainstorming,IdeaListingand Grouping

S.No	IdeaDescription	Category
1	VisualizeaveragepricebySalePriceBin	PricingInsights
2	Analyzeimpactofnumberofbedroomsonsale price	PropertyFeatures
3	Explorerepresentation betweenTotalAreaandPrice(scatterplot)	Size-BasedPricing
4	Compareprices forrenovatedvs.non-renovated homes	RenovationAnalysis
5	Groupinsights byZipcode Clusters	GeographicalComparison
6	Analyzehouseconditionvs. priceusingdummyvariables	Quality-BasedPricing
7	Addcalculatedfield:TotalAreaSqft	DataPreparation
8	CreateSalePriceBinwith100k intervals	Binning/Categorization
9	UseTableaudashboardtocombine insights	DashboardDesign
10	BuildaStoryinTableaufornarrative	Storytelling&Reporting
11	EmbedDashboardinWebApplicationusingFlask	Deployment&Integration
12	AddfiltersforBedrooms,Condition,RenovationinDashboard	InteractiveExploration

Step 3:IdeaPrioritizationTable

S.No	IdeaDescription	Impact	Feasibility	Priority
1	Visualizeaverage sale price by Sale Price Bin	High	Easy	High
2	Analyze impact of number of bedrooms on sale price	High	Easy	High
3	Explore Total Area vs Price (scatter plot)	High	Easy	High
4	Compare prices for renovated vs. non-renovated homes	High	Medium	High
5	Group insights by Zipcode Clusters	Medium	Medium	Medium
6	Analyze house condition vs. price	High	Medium	High
7	Add calculated field: Total Area Sqft	Medium	Easy	High
8	Create Sale Price Bin with 100k intervals	Medium	Easy	High
9	Use Tableau dashboard to combine insights	High	Easy	High
10	Build a Story in Tableau	High	Medium	High
11	Embed Dashboard in Web Application	High	Hard	Medium
12	Add filters for Bedrooms, Condition, Renovation	Medium	Easy	Medium

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

25 minutes

Idea	Importance	Feasibility	Notes
1. Renovation vs Sale Price Dashboard	☆ High	☑ High	Directly connects to business decisions and is easy to visualize in Tableau.
2. Bathroom/ Floor/Age Comparison Chart	☆ High	☑ Medium	Valuable pattern insight; requires grouping and combining features.
3. Add Filters (year, price, features) in Tableau	☆ High	☑ High	Makes dashboards dynamic; easily implemented using Tableau filters.
4. Add Tooltips for Data Insight	☆ Medium	☑ High	Improves user understanding; quick to add in Tableau.
5. Forecast Future Pricing Trends	☆ High	⚡ Low	Valuable, but needs time-series modeling; more complex.
6. Create Storyboards for Presentations	☆ Medium	☑ Medium	Good for communication; depends on team's design ability.
7. Feature Heatmap by Region	⚡ Medium	⚡ Low	Interesting, but requires geospatial data and custom visuals.

After you collaborate

You can export the mural as an image or pdf to share with members of your company who might find it helpful.

Quick edit-ons

- Share the mural**
Share a new link to the mural with stakeholders to keep them in the loop about the outcomes of the session.
- Export the mural**
Export a copy of the mural as a PNG or PDF to attach to emails, include in slides, or save in your drive.

Renovation vs Sale Price Dashboard

House Feature Comparison (Bathrooms, Floors, Age)

Interactive Filters & Tooltips in Tableau

Forecasting Future Trends (optional based on time)

High-Value (Plan if time/resources allow):

- Forecast pricing trends using Tableau's time-series features

Low-Hanging Fruit (Nice to have):

- Create storyboards for internal use or presentations

Avoid for Now:

- Regional heatmaps (require more geospatial data)
- Complex predictions without clear historical data

3. Requirement analysis

Customer Journey map

Customer Journey Map: Housing Market Trends Dashboard

Stage	Actions & Touchpoints	Experience & Emotions	Pain Points	Opportunities	User Goals
Awareness	- Sees dashboard via social media, newsletter, Tableau Public - Reads title/summary	Curious, Interested	Unclear if dashboard is relevant	Use benefit-driven titles, visual thumbnails	Attract interest and clarify purpose
Consideration	- Clicks dashboard link - Reads introduction, explores layout	Engaged, Cautious	Overwhelmed by layout, unsure where to start	Add guided walkthrough, simplify navigation	Understand the dashboard and its features
Exploration	- Uses filters for location, price, features - Views charts (bar, scatter, pie, etc)	Excited, Inquisitive	Filters not intuitive, charts slow to load	Add example queries, improve speed	Discover valuable insights
Decision	- Exports visuals - Shares dashboard - Bookmarks or downloads insights	Satisfied, Confident	Limited export options or unclear formats	Enable easy download/share, offer export guides	Preserve and share findings
Retention	- Subscribes for updates - Revisits for new data - Leaves feedback	Loyal, Empowered	No update notifications, feedback unacknowledged	Enable email updates, actively respond to feedback	Stay informed and engaged

Solution Requirement

Functional Requirements:

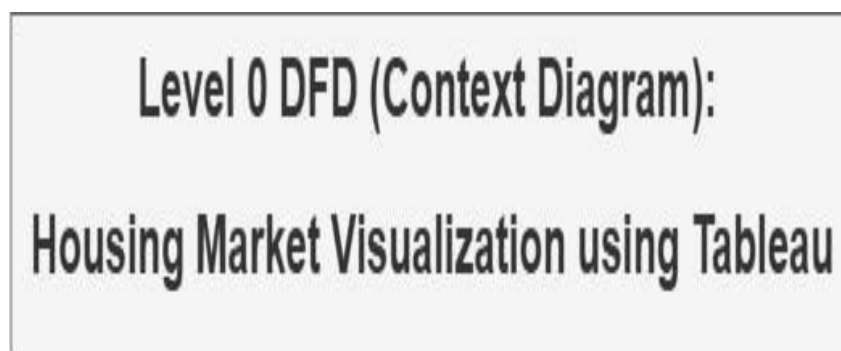
FR No.	Functional Requirement (Epic)	SubRequirement (Story/Sub-Task)
FR-1	Data Upload	Upload dataset (CSV/Excel) Validate file type/format
FR-2	Data Cleaning & Preparation	Handle missing/null values Create calculated fields
FR-3	Visualization	Generate charts: house age, renovation, features
FR-4	Dashboard Interaction	Add filters (year, bedrooms) Allow download/export
FR-5	Dashboard Testing	Test load time and performance Verify filter responsiveness
FR-6	Dashboard Publishing	Publish on Tableau Public Enable embed/share features

Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard must be user-friendly, with clear navigation and interactive filters
NFR-2	Security	Uploaded data must be protected from unauthorized access
NFR-3	Reliability	System should display accurate and consistent visualizations
NFR-4	Performance	Dashboard should load within 2 seconds with filters applied
NFR-5	Availability	Dashboard should be accessible online 24/7 via Tableau Public
NFR-6	Scalability	The system must support future data expansion or new visualization requirements

DataFlowDiagram

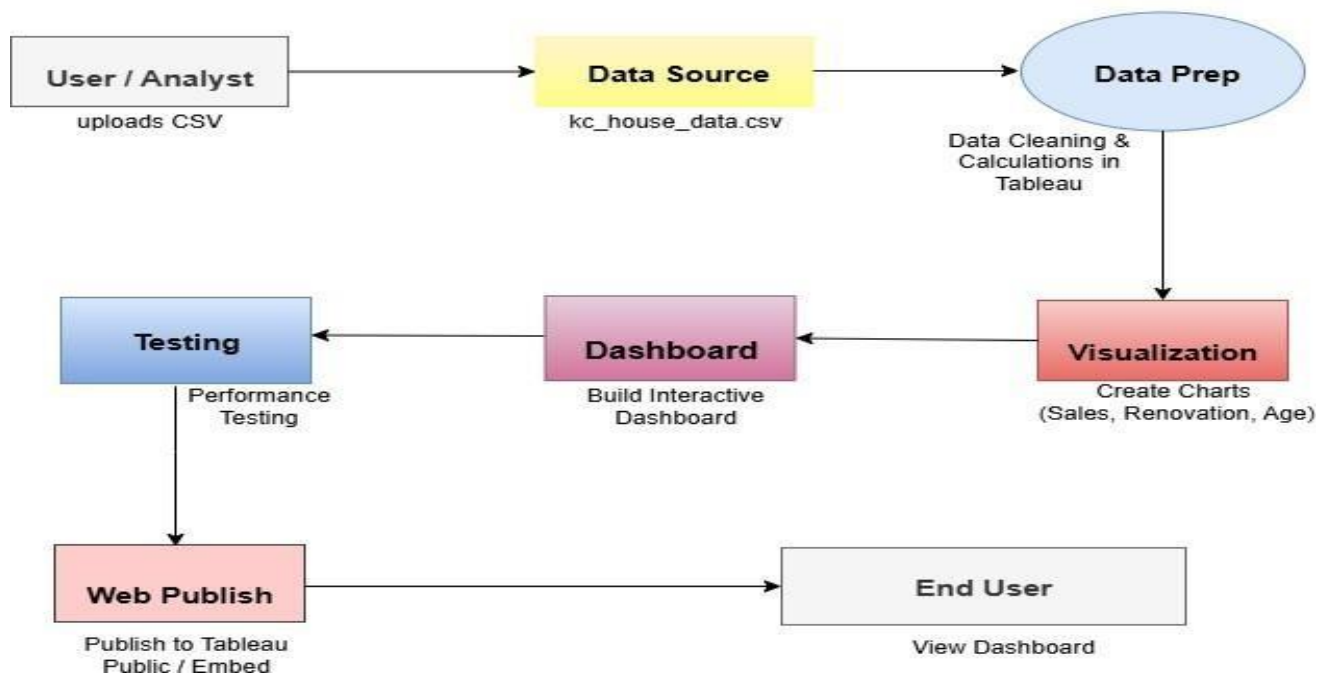
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



DFD Level 0 – Context Diagram Flow

- User/Analyst uploads housing data (CSV/Excel) into the visualization system.
- System processes the data and creates meaningful visualizations.
- End users (executives, analysts) view insights via an interactive Tableau dashboard.

Level 1 DFD (Expanded): Housing Market Visualization using Tableau



DFD Level 1 – Expanded Flow

- File upload and validation (check file type, structure).
- Handle missing/null values if present.
- Clean the data (e.g., format dates).
- Create calculated fields:
e.g., House Age, Years Since Renovation, Price/Sqft.
- Create charts using Tableau: Total Sales, House Age, Bedrooms/Floors comparison, etc.
- Combine visuals into a filtered interactive dashboard (with dropdowns/sliders).
- Load performance, filter functionality, field calculation checks.
- Publish to Tableau Public
- Embed into a website / share with stakeholders

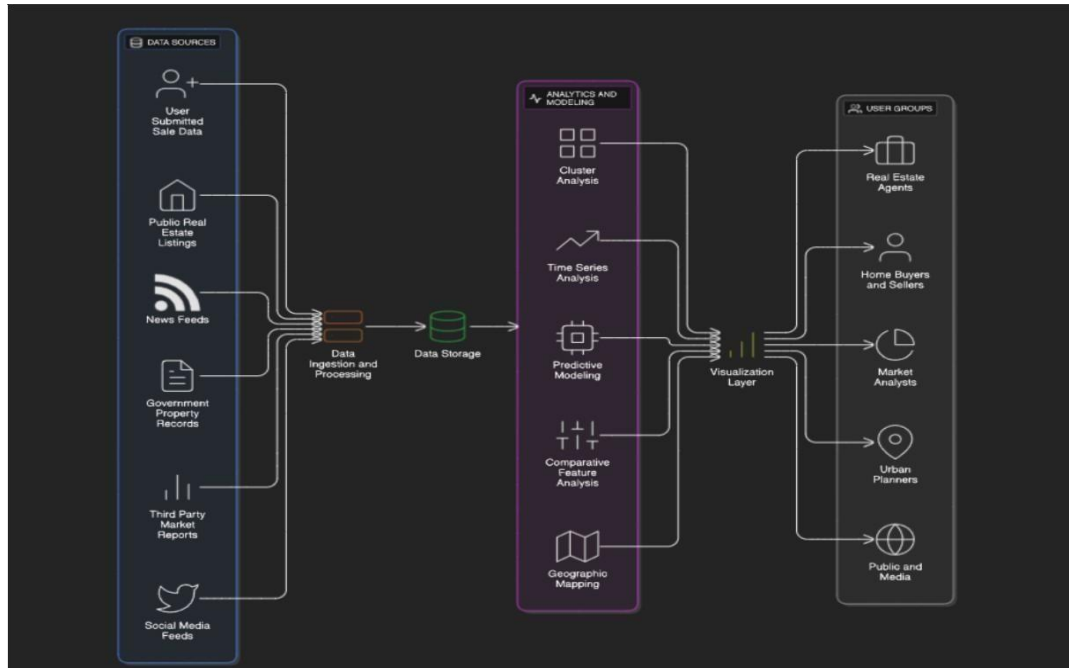
User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story/Task	Acceptance criteria	Priority	Release
Data Analyst	Data Upload	USN-1	As a data analyst, I can upload the housing dataset to begin the visualization process.	Dataset is successfully uploaded and read by Tableau.	High	Sprint-1
Data Analyst	Data Preparation	USN-2	As a data analyst, I can clean and transform the dataset in Tableau.	Null values removed, calculated fields added (e.g., House Age).	High	Sprint-1
Data Analyst	Visualization Creation	USN-3	As a data analyst, I can create charts based on renovation, house age, and features.	Visuals like bar/pie charts are created and reflect accurate data.	High	Sprint-1
Viewer	View Dashboard	USN-4	As a viewer, I can interact with the dashboard using filters.	Filters (e.g., Year, Region, Bedrooms) function correctly.	High	Sprint-2
Executive	Dashboard Export	USN-5	As an executive, I can export dashboard visuals for reports.	PDF/image exports are generated from Tableau dashboard.	Medium	Sprint-2
Admin	Dashboard Sharing	USN-6	As an admin, I can publish or embed the dashboard on a website.	Dashboard appears embedded and accessible to stakeholders.	Medium	Sprint-2

Priority	Release
High	Sprint-1
High	Sprint-1
High	Sprint-1
High	Sprint-2
Medium	Sprint-2
Medium	Sprint-2

Technology Stack



Components&Technologies

S.No	Component	Description	Technology
1	User Interface	Web access to visual dashboards	Web UI using Tableau Public
2	Application Logic-1	Data pre-cleaning and transformation	Python (Pandas, NumPy)
3	Application Logic-2	Dashboard logic, filters, calculated fields	Tableau Desktop
4	Application Logic-3	Interactive story design	Tableau Story Builder
5	Database	Structured data in tabular format	CSV file (Transformed Housing Data)
6	Cloud Database	Optional sync with Google Sheets or Tableau Online	Google Sheets / Tableau Cloud
7	File Storage	Storing housing datasets	Google Drive / Local File System
8	External API-1	(Not used in current scope)	N/A
9	External API-2	(Not used in current scope)	N/A
10	Machine Learning Model	Future enhancement for price prediction	Python Regression Model (optional)
11	Infrastructure	Deployment & Hosting	Local System, Tableau Public (Cloud)

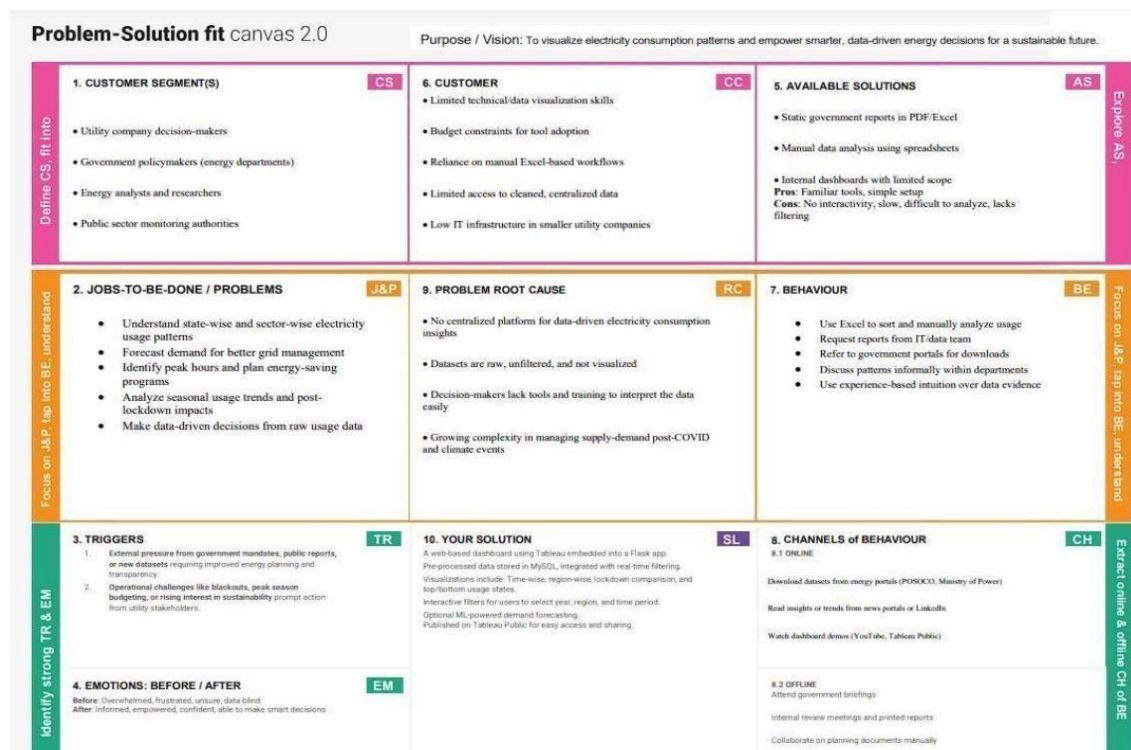
4. Projectdesign

ProblemSolutionFit

The Problem-SolutionFit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.



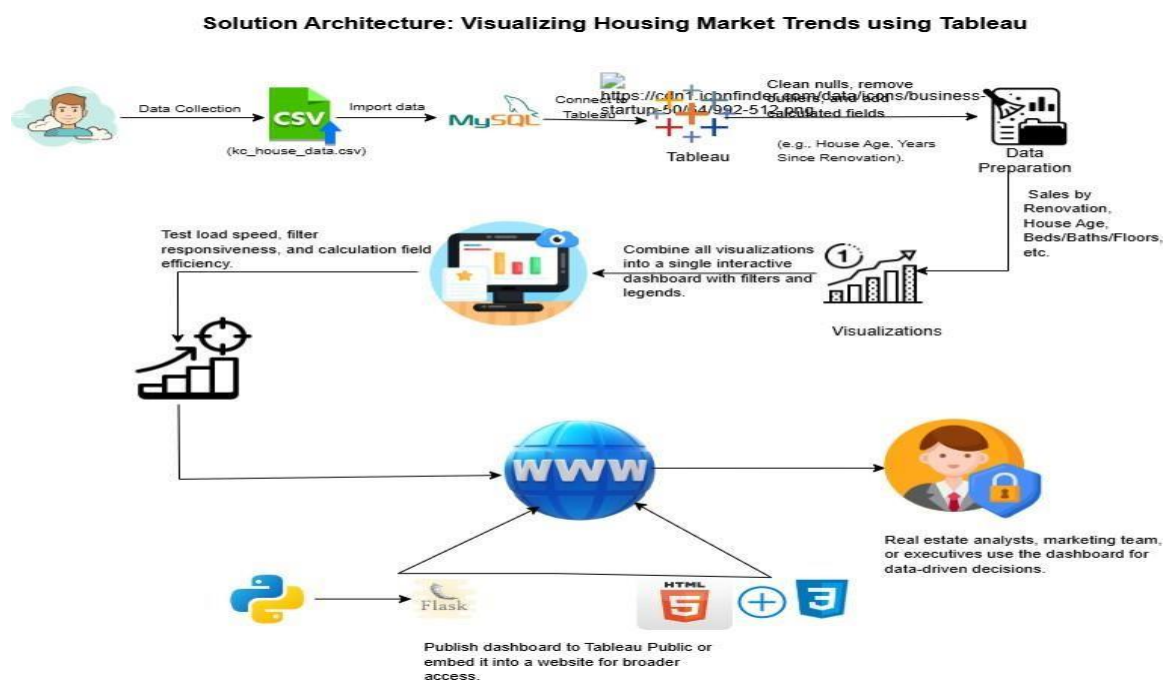
ProposedSolutionTemplate

S.No.	Parameter	Description
1.	ProblemStatement(Problemto be solved)	Identifying the housing features—such as renovationage,totalarea,andhousecondition—thatsignificantlyinfluencesalepricesisdifficult due to unstructured, raw real estate data. Most users lack tools or skills to interpret this data effectively.
2.	Idea/Solutiondescription	AninteractiveTableaudashboardthatanalyzes housingdataandvisualizestrendsbasedonkey features like sale price, number of bedrooms, renovations, and floor area. Users can filter, compare,andexploreinsightsinreal-time.
3.	Novelty/Uniqueness	Unlike static listings or Excel reports, this solution offers interactive, visual exploration of housingmarkettrendsusingTableau.Itcombines spatial, temporal, and feature-based analysis for deeperinsight.
4.	SocialImpact/CustomerSatisfaction	Helpshomebuyersandinvestorsmakeinformed decisions. Enhances data transparency in the housingmarket,reduce[s]agentdependency,and support[s]digitalliteracyinrealestateanalysis.
5.	BusinessModel(RevenueModel)	Canbe monetized byoffering it to realestate companies, property consultants, or data platform[s]asaSaaS[ool]orsubscription-based dashboardservice.
6.	ScalabilityoftheSolution	Highlyscalabletomultiplecitiesorcountriesby plugging in local housing datasets or APIs. The modelisreusableacrossvariousgeographiesand datasources.

4.2 Solution Architecture

Solution architecture is a complex process—with many sub-processes—that bridges the gap between business problems and technology solutions. Its goals are to:

- The architecture separates data preprocessing, storage, visualization, and UI layers—making it easy to maintain, scale, and enhance.
- Cleaned data from MySQL is visualized using Tableau dashboards, offering region-wise, year-wise, and seasonal insights with filtering capabilities.
- Dashboards are embedded into a Flask-based web interface, allowing end users to interact with visual data through a user-friendly portal.
- The solution supports future extensions like forecasting models and can be deployed locally or on cloud platforms like Heroku or AWS.



5. Project planning & scheduling

5.1 Project Planning

UserStory /Task	Poin	Priority	AssignedTo
Sprin1			
DataSetup–USN-1	3	High	DolaGowthami
DataCleaning–USN-2	4	High	AnnamdevulaHema
FieldCreation– USN-3	2	Medium	DolaGowthami
PriceBinning –USN-4	2	Medium	AnnamdevulaHema

UserStory /Task	Poin	Priority	AssignedTo
USN-5–Createcharts: pricevsfeatures	5	High	DolaGowthami

UserStory /Task	Poin	Priority	AssignedTo
USN-6 – Build interactive Tableaudashboardwithfilters	3	High	AnnamdevulaHema
USN-7–Stylethedashboardfor readability	2	Medium	DolaGowthami

Sprin3: Styling & Integration

User Story / Task	Poin	Priority	AssignedTo
USN-8–TableauStorytelling (step-by-step insights)	2	Medium	AnnamdevulaHema
USN-9–EmbedTableau dashboard into Flask	4	High	DolaGowthami
USN-10–UITestingfor embedded dashboard	2	Medium	AnnamdevulaHema
USN-11–Finalproject documentation	3	High	DolaGowthami

UserStory /Task	Poin	Priority	AssignedTo
USN-12–Demopreparationand rehearsal	2	Medium	Both
USN-13–BugfixingandQA	2	Medium	Both

Project Tracker, Velocity & Burndown Chart

Sprint	TotalStoryPoints	Duration	StartDate	EndDate	PointsCompleted	ReleaseDate
Sprint1	11	4Days	20June2025	23June2025	11	27June2025
Sprint2	8	4Days	24June2025	27June2025	10	1July2025
Sprint3	10	4Days	28June2025	01July 2025	7	5July2025
Sprint4	6	4Days	02July2025	05 July 2025	7	9July2025

VelocityChartSummary

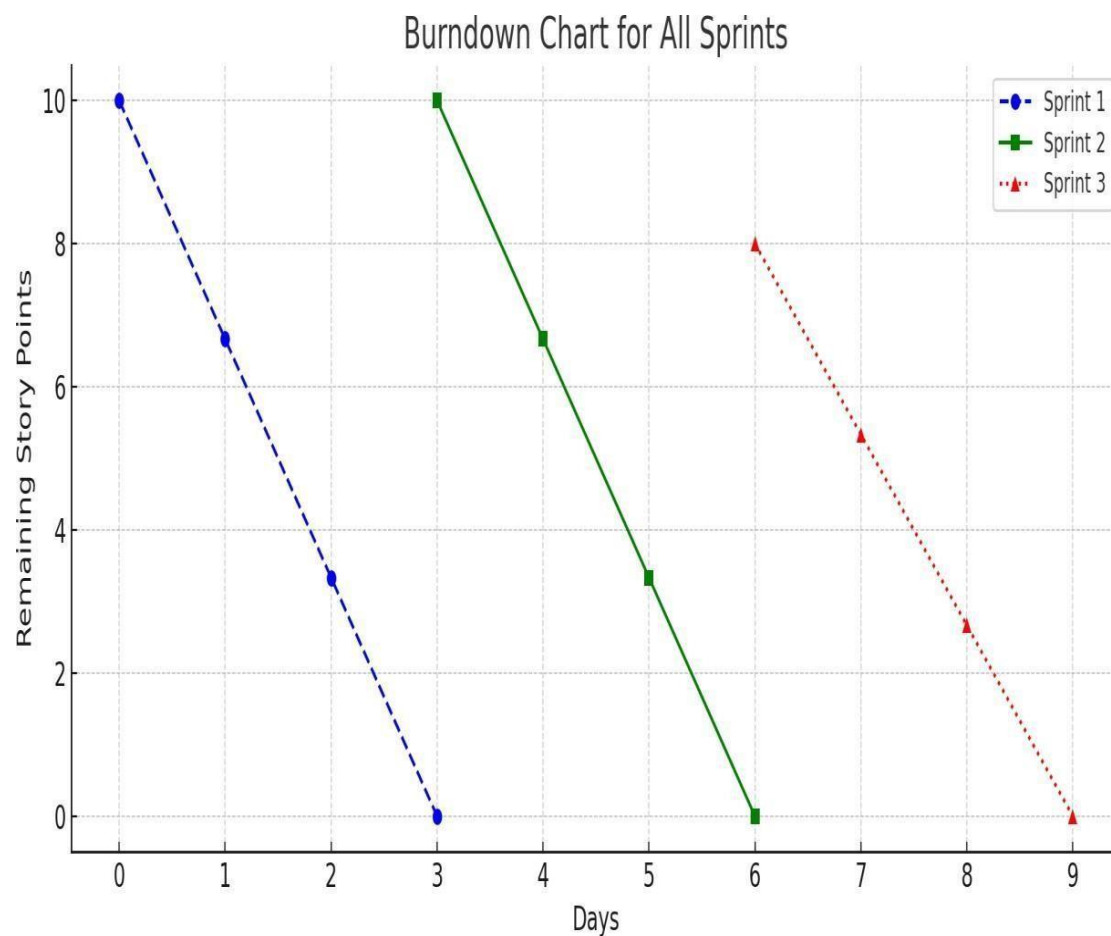
- **Velocity(CompletedPoints/Sprint):**
 - Sprint1:9
 - Sprint2:6
 - Sprint3:7
 - **AverageVelocity(first3sprints):**
 $AV=28/9 \approx 3.11$ storypoints/day
-

BurndownChartData

Assumetotalbacklog=35StoryPoints

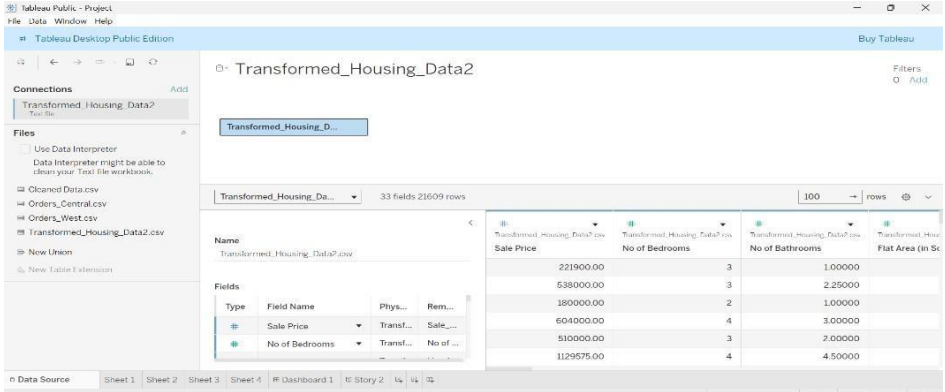
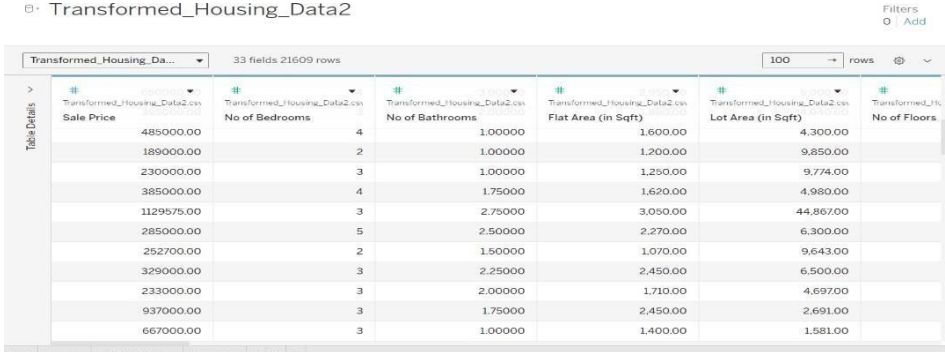
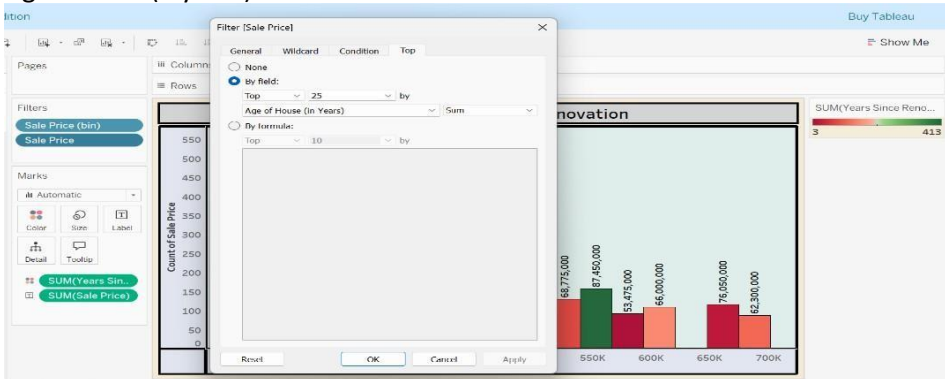
SprintRemainingPoints(Start)CompletedRemainingPoints(End)

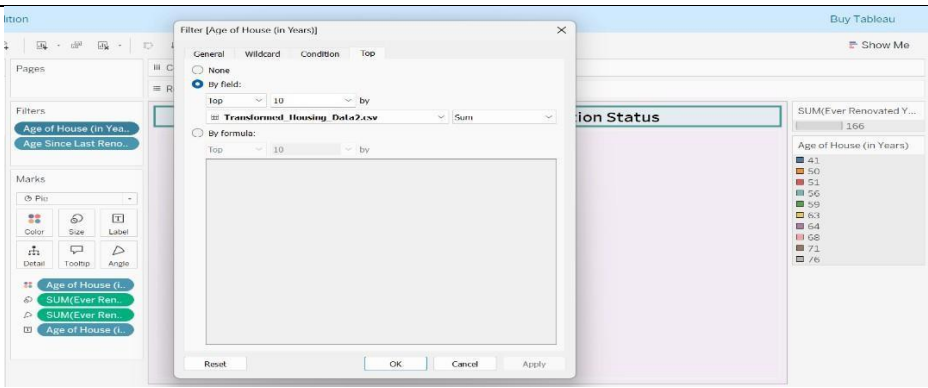
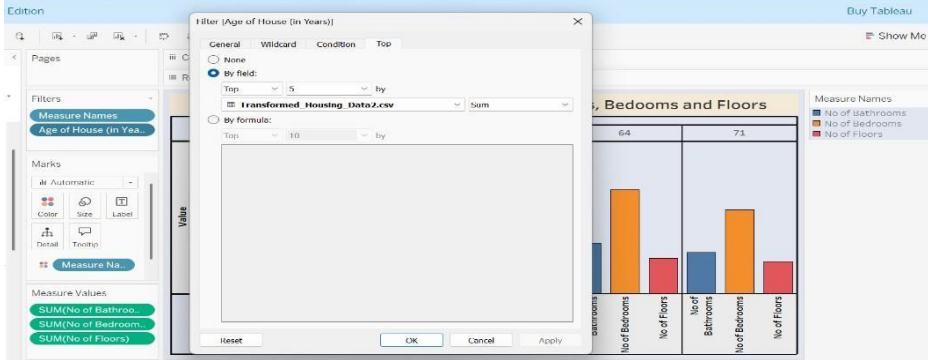
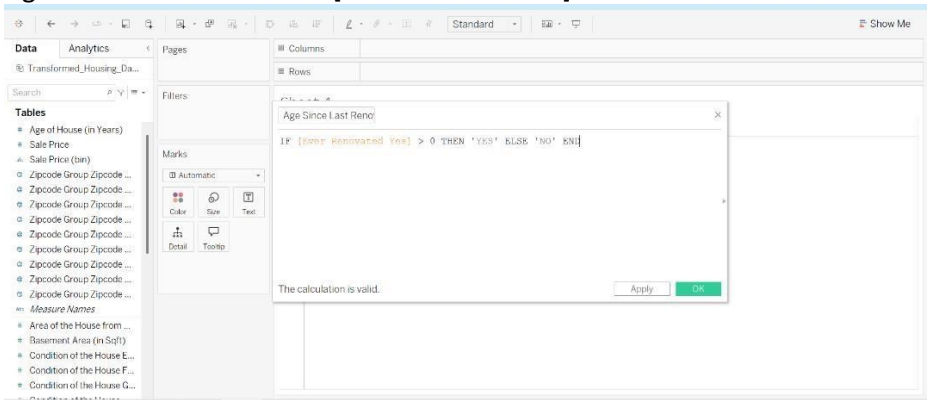
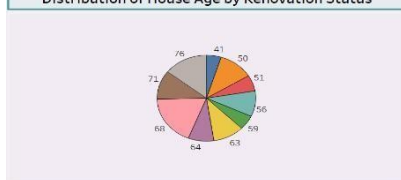
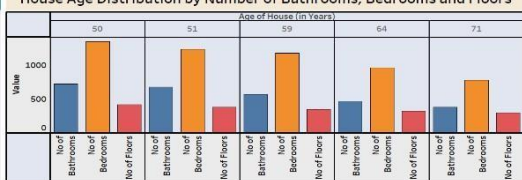
Sprint1	35	9	26
Sprint2	26	6	20
Sprint3	20	7	13
Sprint4	13	TBD	TBD



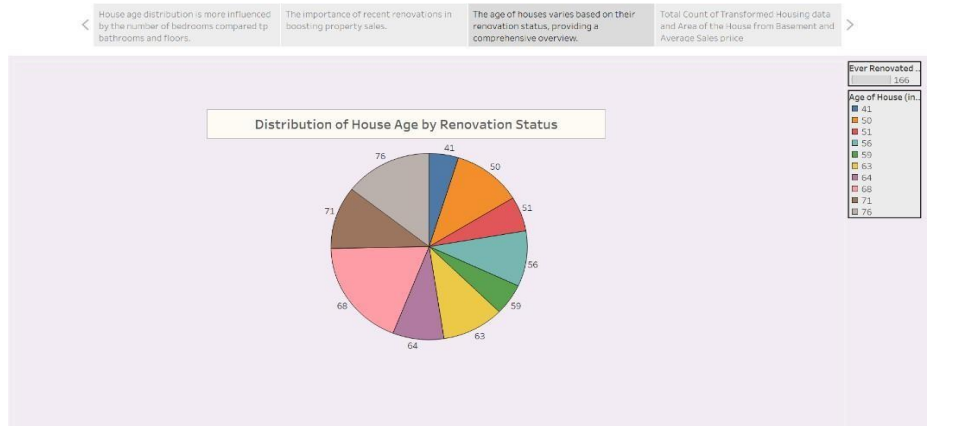
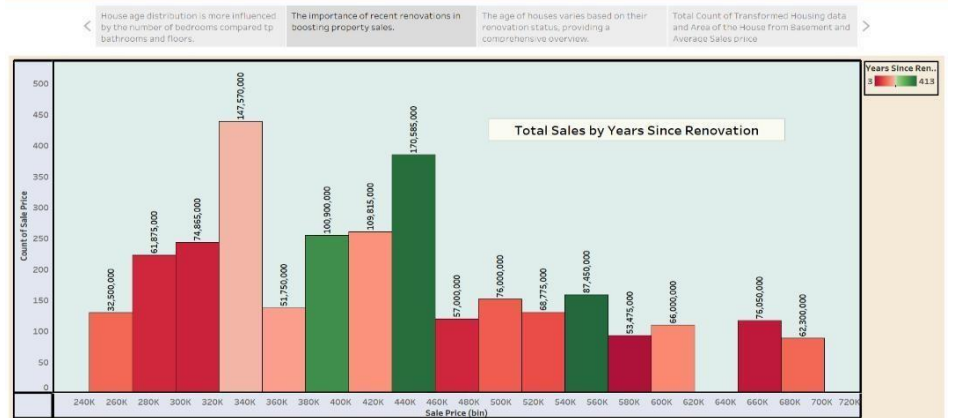
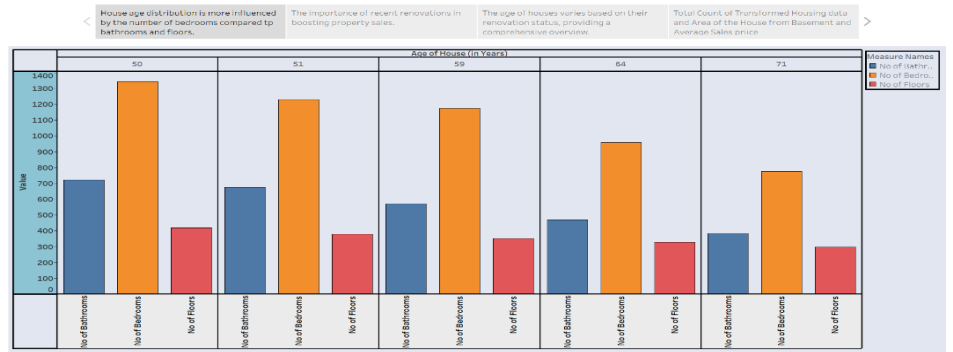
6. Functional and performance testing

6.1 Performance Testing

S.No	Parameter	Screenshot/Values
1.	Data Rendered	Housing data with 21 columns and 2000+ records loaded from CSV into Tableau. 
2.	Data Preprocessing	- Removed null values from 'price' and 'renovation date' 
3.	Utilization of Filters	Filters applied on: Sales Price Age of House (in years) 

								
								
4.	Calculationfields Used	AgeSinceLastRenovation=IF[EverRenovatedYes]>0THEN‘YES’ELSE‘NO’ END 						
5.	Dashboarddesign	NoofVisualizations/Graphs– Total Sales By Year Since Renovation DistributionofHouseAgebyRenovationStatus HouseAgeDistributionbyNumberofBathrooms,BedroomsandFloors Comprehensive Home Price Analysis <table><tr><td>Count of Transformed_Housing_Data2.csv</td><td>21,609</td></tr><tr><td>Area of the House from Basement (in Sqft)</td><td>38,643,798</td></tr><tr><td>Avg. Sale Price</td><td>511,619</td></tr></table> Distribution of House Age by Renovation Status  House Age Distribution by Number of Bathrooms, Bedrooms and Floors 	Count of Transformed_Housing_Data2.csv	21,609	Area of the House from Basement (in Sqft)	38,643,798	Avg. Sale Price	511,619
Count of Transformed_Housing_Data2.csv	21,609							
Area of the House from Basement (in Sqft)	38,643,798							
Avg. Sale Price	511,619							

NoofVisualizations/Graphs– Start include 4 Sheets.



House age distribution is more influenced by the number of bedrooms compared to bathrooms and floors.

The importance of recent renovations in boosting property sales.

The age of houses varies based on their renovation status, providing a comprehensive overview.

Total Count of Transformed Housing data and Area of the House from Basement and Average Sales price.

Count of Transformed_Housing_Data2.csv	21,609
Area of the House from Basement (in Sqft)	38,643,798
Avg. Sale Price	511,619

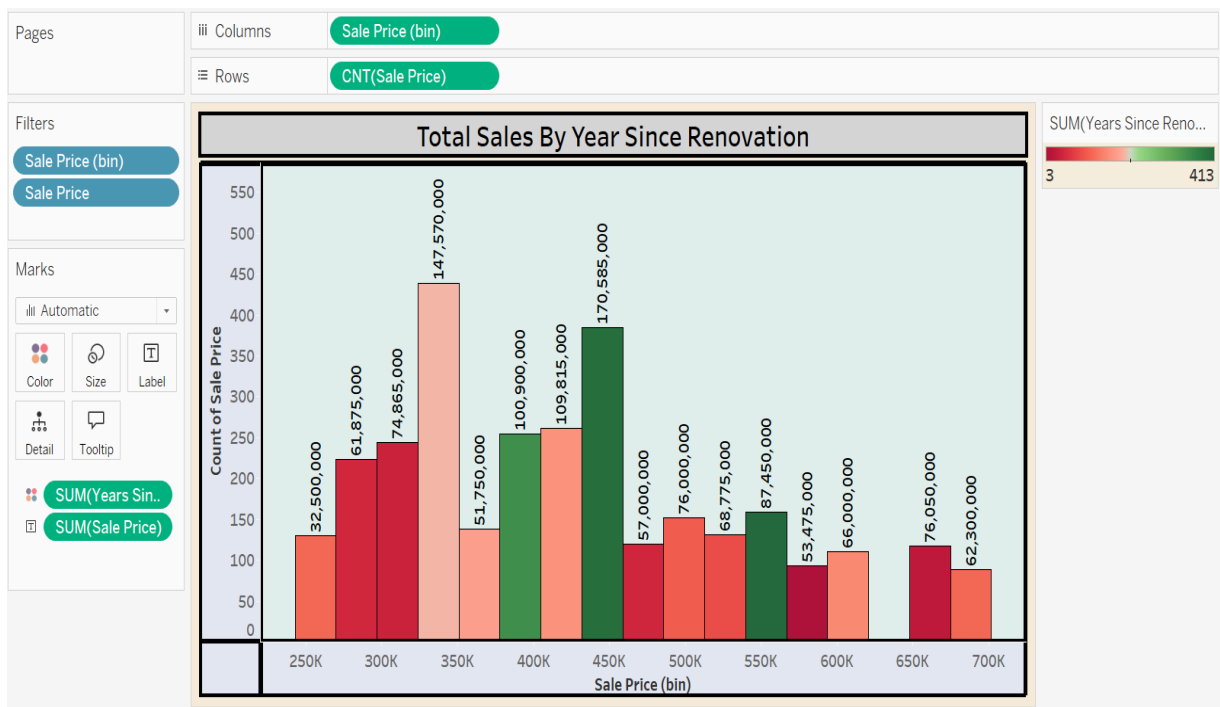
7. Results

7.1 Output Screenshots

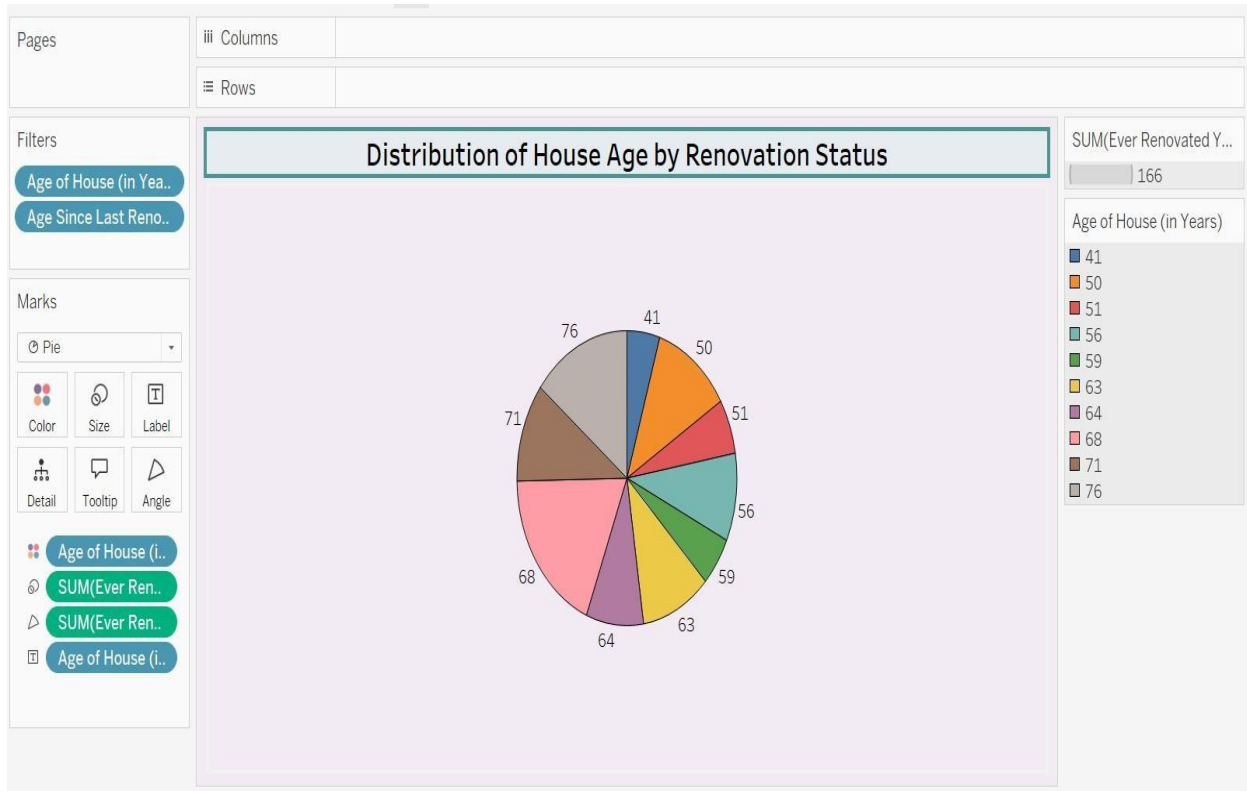
Output of Sheet 1

Pages	Columns	
	Rows	Measure Names
Filters		
Measure Names		
Marks		
Automatic		
Color		
Size		
Text		
Detail		
Tooltip		
Measure ..		
Measure Val..		
Measure Values		
CNT(Transformed_Ho..		
SUM(Area of the Ho..		
AVG(Sale Price)		
	Count of Transformed_Housing_Data2.csv	21,609
	Area of the House from Basement (in Sqft)	38,643,798
	Avg. Sale Price	511,619

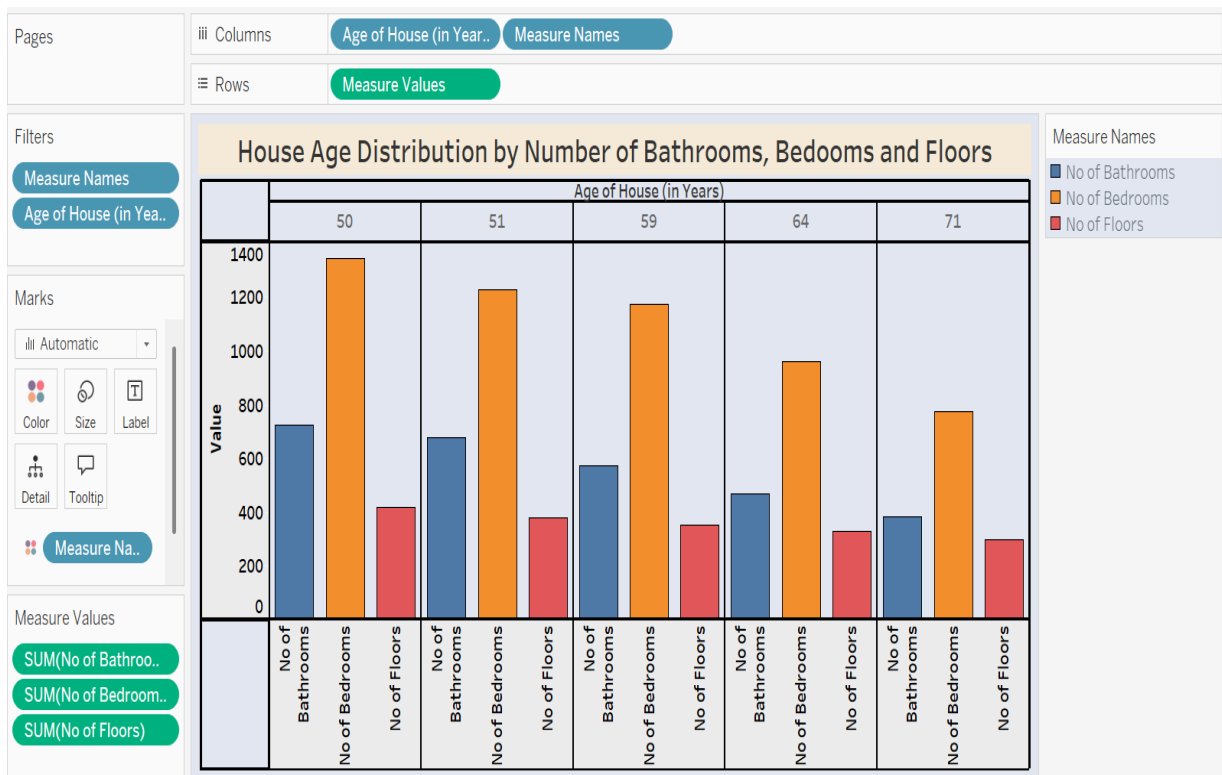
Output of Sheet 2



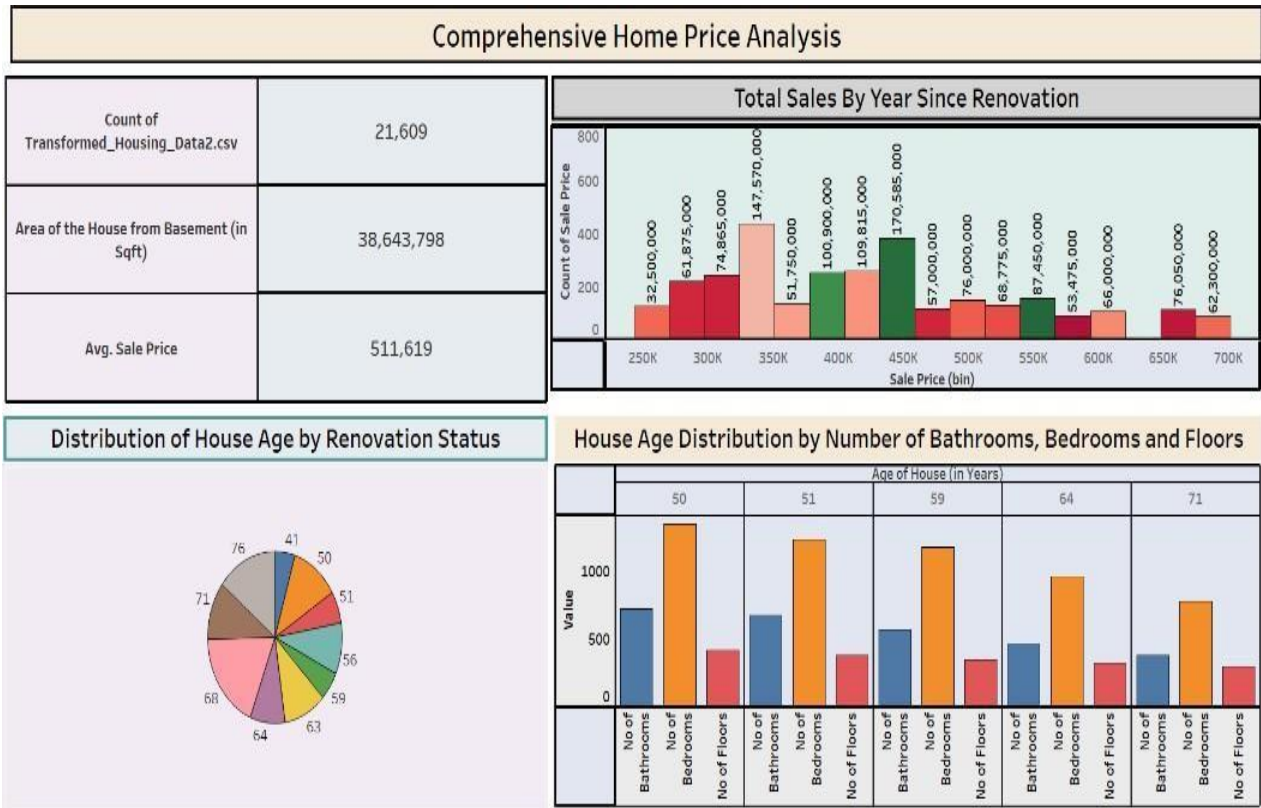
Output of Sheet 3



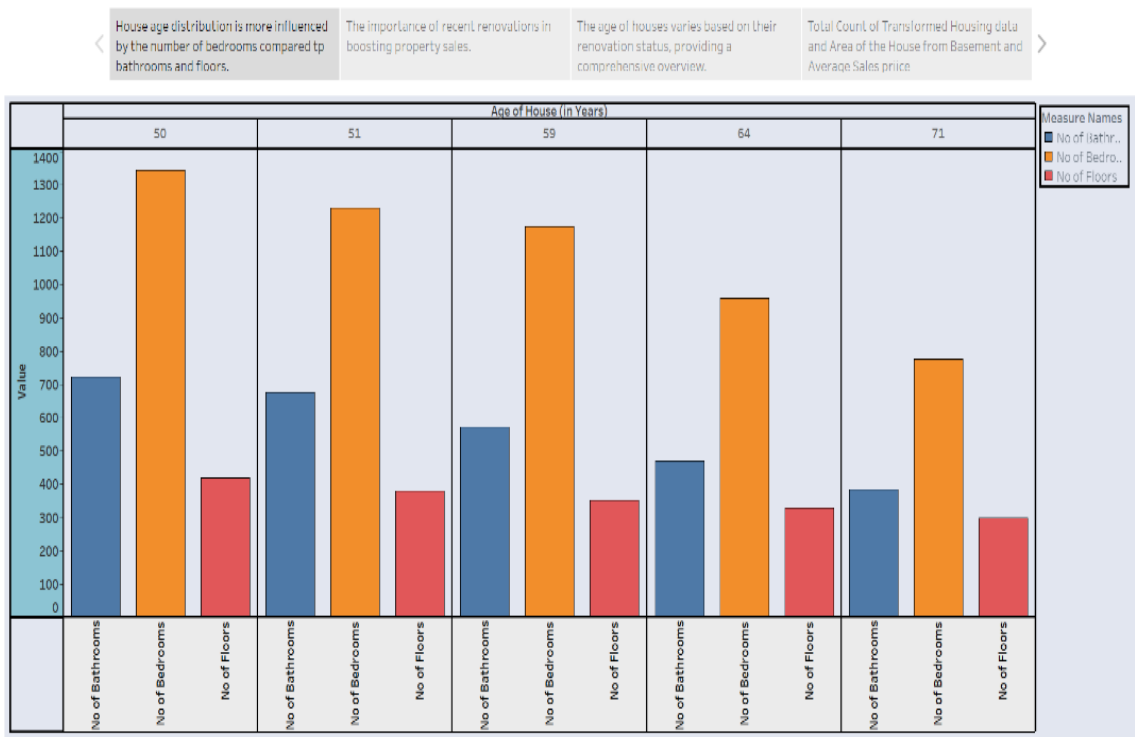
Output of Sheet 4



OutputofDashboard



OutputofStory

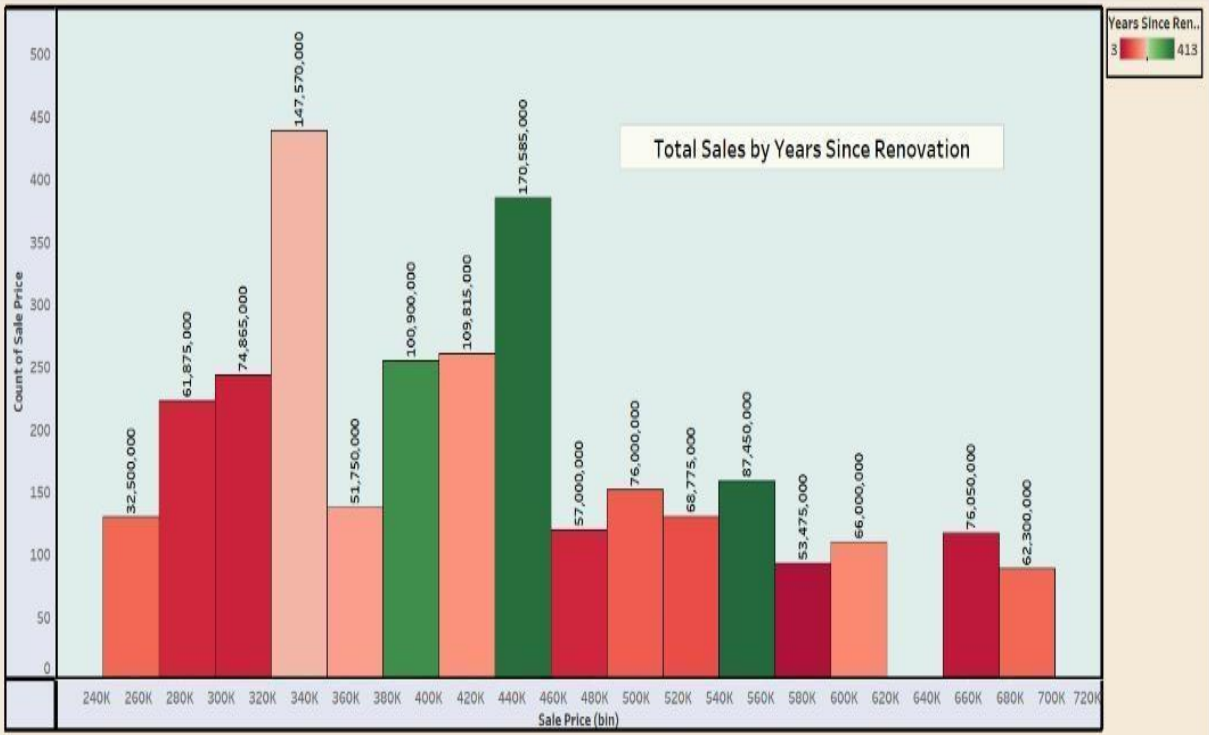


House age distribution is more influenced by the number of bedrooms compared tp bathrooms and floors.

The importance of recent renovations in boosting property sales.

The age of houses varies based on their renovation status, providing a comprehensive overview.

Total Count of Transformed Housing data and Area of the House from Basement and Average Sales price

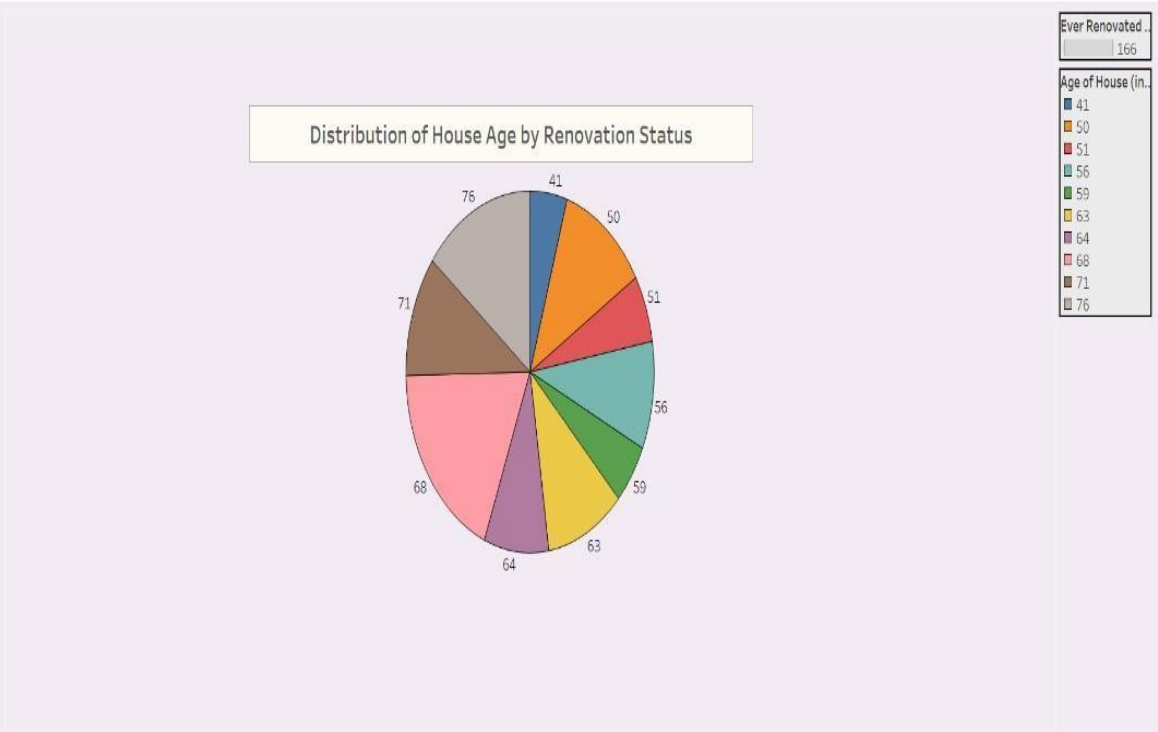


House age distribution is more influenced by the number of bedrooms compared tp bathrooms and floors.

The importance of recent renovations in boosting property sales.

The age of houses varies based on their renovation status, providing a comprehensive overview.

Total Count of Transformed Housing data and Area of the House from Basement and Average Sales price



House age distribution is more influenced by the number of bedrooms compared tp. bathrooms and floors. The importance of recent renovations in boosting property sales. The age of houses varies based on their renovation status, providing a comprehensive overview. Total Count of Transformed Housing data and Area of the House from Basement and Average Sales price	
Count of Transformed_Housing_Data2.csv	21,609
Area of the House from Basement (in Sqft)	38,643,798
Avg. Sale Price	511,619

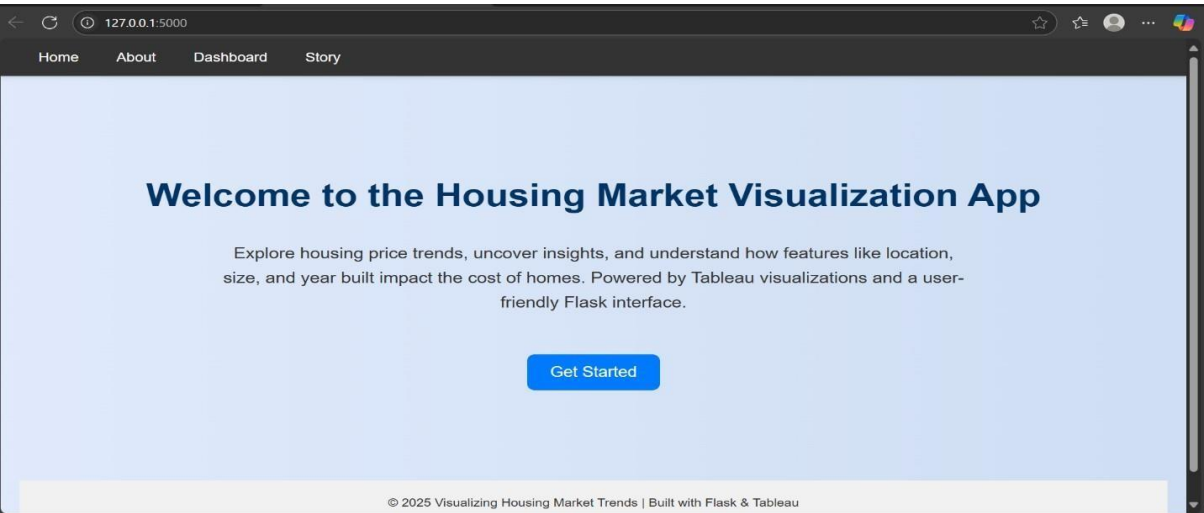
TableaupublicDashboard link:

https://public.tableau.com/app/profile/gowthami.dola/viz/Project_17512069507510/Dashboard1?publish=yes

TableaupublicStorylink:

<https://public.tableau.com/app/profile/gowthami.dola/viz/HousetrendsStory/HouseStory?publish=yes>

OutputofWebSite:



Home
About
Dashboard
Story

About the Project

Project Title: *Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau*

This project focuses on uncovering key insights into the housing market by analyzing sale prices in relation to various features such as location, square footage, number of bedrooms/bathrooms, and year built.

Using **Tableau**, we have built interactive visualizations and dashboards that help users explore and understand trends in the housing market over time. The data is visually represented through charts, graphs, maps, and storytelling dashboards that make the analysis intuitive and insightful.

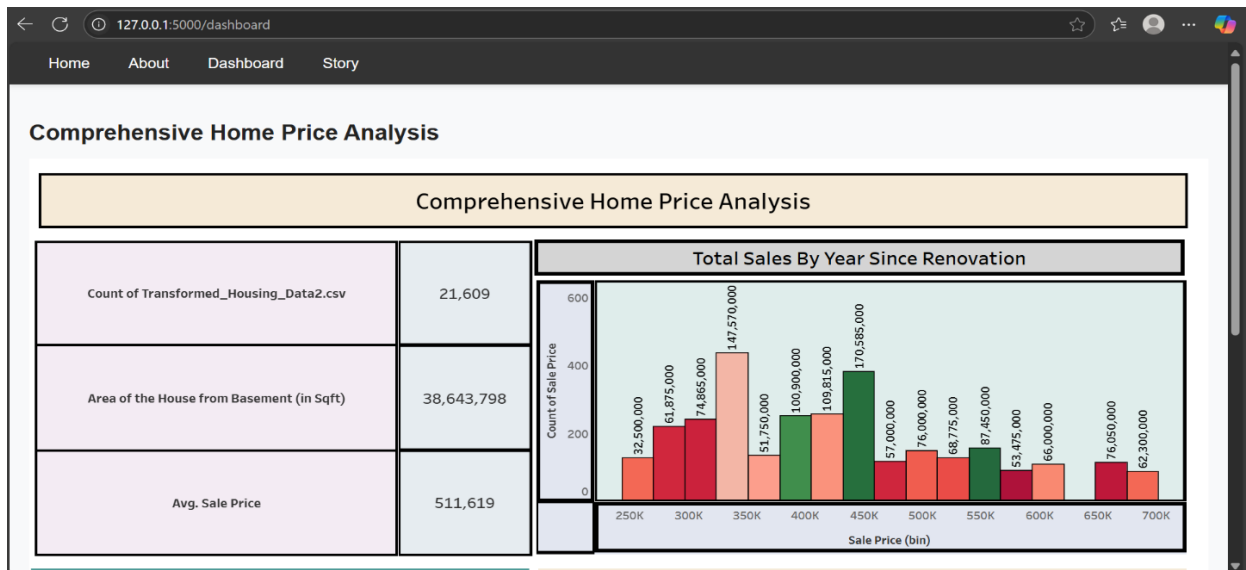
Objectives

- To identify the major factors influencing house prices.
- To visualize how house prices vary by region and over time.
- To enable users to interactively explore the data and draw conclusions.

Tools & Technologies Used

- Flask** – Python framework used to build the web application.
- Tableau Public** – Used to create and host the dashboards and storyboards.
- HTML/CSS** – For building the front-end user interface.

Why Tableau?



8. Advantages&disadvantages

■ AdvantagesofCustomerJourneyMapping

1. **User-CentricUnderstanding**
Helpsyoustepintothe customer'sshoesandunderstandtheirneeds,painpoints,and expectations.
2. **ImprovedExperienceDesign**
Enablesyoutodesigndashboardsandinsightsthatareintuitiveandalignedwithrealuser behaviour.
3. **IdentifiesGaps&Opportunities**
Clearlyshowswhichstepsinthejourneyarefrustratingorinefficient,guiding improvement.
4. **Cross-FunctionalAlignment**
Facilitatesbettercommunicationbetweendevelopers,analysts,marketing,andend-users.
5. **Data-DrivenDecisions**
Supportsbetterprioritizationoffeaturesorcontentthat improvetheoveralluserjourney.
6. **IncreasedUserSatisfaction**
Helpsensuresmoother andmoresatisfyinginteractionswiththedashboardandinsights.

+DisadvantagesofCustomerJourneyMapping

1. **Time-ConsumingtoCreate**
Mappingeachjourneyindetail(especiallyformultiplepersonas)takestimeand resources.
2. **RequiresRealUserResearch**
Withoutrealuserinterviewsorobservations,assumptionscanleadtomisleading conclusions.
3. **ComplexforMultipleJourneys**
Differentusertypesmayrequireseparate maps,making managementharder.
4. **MayOversimplify Reality**
ACJM isamodel—itmightnotcaptureeveryedgecaseor complexpath.
5. **NotAlwaysActionedUpon**
InsightsfromCJMsmaybeignoredifthere'sno follow-throughbyteamsordecision-makers.
6. **NeedsRegularUpdating**
Customerbehavioursandtechnologyevolve—yourjourneymapcanquicklybecome outdated.

9. Conclusion:

The project effectively utilized Tableau to visualize and analyze housing market data, focusing on key factors such as renovation history, house age, number of bedrooms/bathrooms, and sale prices. Through interactive dashboards, we uncovered trends and patterns that support strategic decision-making for stakeholders like real estate analysts and company executives.

The visualizations provided clear insights into:

- The impact of renovations on house prices
- Preferences based on house features like bathrooms and floors
- Distribution of house age and its relationship to sale trends
- Geographic segmentation through zip code groupings

By transforming raw data into intuitive dashboards, the project bridges the gap between complex housing data and actionable business intelligence.

Future Scope

While the current project offers meaningful insights, there is significant scope for future enhancements:

1. **Integration of Real-Time Data**
 - Connect dashboards to a live data source (e.g., via APIs or Google Sheets) to keep insights up-to-date.
2. **Predictive Analytics with Machine Learning**
 - Use regression or classification models to forecast house prices based on features.
3. **Enhanced Geographic Visualizations**
 - Incorporate heatmaps or interactive maps based on latitude/longitude or zip codes.
4. **User Personalization**
 - Allow users to save filters or generate custom reports based on their use case.
5. **Mobile Dashboard Optimization**
 - Make visualizations more responsive and accessible on mobile devices for real-time field access.
6. **Integration with CRM/ERP Systems**
 - Embed Tableau dashboards into business systems to align insights with operations.
7. **Advanced Filtering & Drill-Down Capabilities**
 - Enable multi-layered filtering (e.g., filter by renovation, then by bathroom count)

10. Appendix

SourceCode:

about.html

```
{%extends"base.html"%}
{%blocktitle%}About{%endblock%}
{%blockcontent%}
<h2>About theProject</h2>

<p>
  <strong>ProjectTitle:</strong><em>VisualizingHousingMarketTrends:AnAnalysisof Sale
  Prices and Features using Tableau</em>
</p>
<p>
  Thisproject focusesonuncoveringkeyinsightsintothehousingmarketbyanalyzingsale prices in
  relation to various features such as location, square footage, number of bedrooms/bathrooms,
  and year built.
</p>
<p>
  Using<strong>Tableau</strong>,wehavebuiltinteractivevisualizationsanddashboards that help
  users explore and understand trends in the housing market over time. The data is visually
  represented through charts, graphs, maps, and storytelling dashboards that make the analysis
  intuitive and insightful.
</p>
<h3>Objectives</h3>
<ul>
  <li>To identify the major factors influencing house prices.</li>
  <li>To visualize how house prices vary by region and over time.</li>
  <li>To enable users to interactively explore the data and draw conclusions.</li>
</ul>
<h3>Tools&TechnologiesUsed</h3>
<ul>
  <li><strong>Flask</strong>–Python framework used to build the web application.</li>
  <li><strong>TableauPublic</strong>–Used to create and host the dashboards and storyboards.</li>
  <li><strong>HTML/CSS</strong>–For building the front-end user interface.</li>
</ul>

<h3>WhyTableau?</h3>
<p>
  Tableau was chosen for its powerful data visualization capabilities and ease of use. It allows for the
  creation of dynamic dashboards and storylines that make the data not only more accessible but
  also more actionable for users and stakeholders.
</p>
<p>
  Through this project, we aim to demonstrate how visual analytics can simplify complex datasets
  and empower better decision-making in real estate and beyond.
</p>
{%endblock%}
```

Base.html:

```
<!DOCTYPEhtml>
<html>
<head>
  <title>{%blocktitle%}MyTableauApp{%endblock%}</title>
  <linkrel="stylesheet"href="{{url_for('static',filename='style.css')}}">
  <style>
    body{
      font-family:Arial,sans-serif; margin:
      0;
    }
    .navbar{
      background-color:#333;
      overflow: hidden;
      display: flex;
    }
    .navbar a {
      color:white;
      padding: 14px 20px;
      text-decoration:none;
      display: block;
    }
    .navbara:hover{
      background-color:#575757;
    }
    .container {
      padding:20px;
    }
    iframe{
      width: 100%;
      height: 80vh;
      border: none;
      margin-top:20px;
    }
  </style>
</head>
<body>
<divclass="navbar">
  <a href="{{url_for('home')}}">Home</a>
  <a href="{{url_for('about')}}">About</a>
  <a href="{{url_for('dashboard')}}">Dashboard</a>
  <a href="{{url_for('story')}}">Story</a>
</div>
<divclass="container">
  {%blockcontent%}{%endblock%}
</div>
</body>
</html>
```

Dashboard.html:

```
{%extends"base.html"%}
{%blocktitle%}Dashboard{%endblock%}
{%blockcontent%}
<h2>ComprehensiveHomePriceAnalysis</h2>
<!--TableauJavaScriptEmbed-->
<divclass='tableauPlaceholder'id='viz1751207840512'style='position:relative'>
  <noscript>
    <a href='#'>
      <img alt='Comprehensive Home Price Analysis
        'src='https://public.tableau.com/static/images/Pr/Project_17512069507510/Dashboard1/1_r
ss.png'
        style='border:none'/>
    </a>
  </noscript>
  <objectclass='tableauViz'style='display:none;'>
    <paramname='host_url'value='https%3A%2F%2Fpublic.tableau.com%2F'>
    <paramname='embed_code_version'value='3'/>
    <paramname='site_root'value='"/>
    <paramname='name'value='Project_17512069507510/Dashboard1'/>
    <paramname='tabs'value='no'/>
    <paramname='toolbar'value='yes'/>
    <param
      name='static_image'value='https://public.tableau.com/static/images/Pr/Project_17512069507
510/Dashboard1/
1.png'/>
    <paramname='animate_transition'value='yes'/>
    <paramname='display_static_image'value='yes'/>
    <paramname='display_spinner'value='yes'/>
    <paramname='display_overlay'value='yes'/>
    <paramname='display_count'value='yes'/>
    <paramname='language'value='en-US'/>
    <paramname='filter'value='publish=yes'/>
  </object>
</div>
<scripttype='text/javascript'>
  vardivElement=document.getElementById('viz1751207840512');
  varvizElement=divElement.getElementsByTagName('object')[0];
  if (divElement.offsetWidth > 800) {
    vizElement.style.width='100%';
    vizElement.style.height=(divElement.offsetWidth *0.75)+'px';
  }elseif(divElement.offsetWidth>500){
    vizElement.style.width = '100%';
    vizElement.style.height=(divElement.offsetWidth *0.75)+'px';
  }else{
    vizElement.style.width = '100%';
    vizElement.style.height ='1277px';
  }
}
```

```
varscriptElement=document.createElement('script');
```

```

scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';vizElement.parentNode.insertBefore(scriptElem
ent, vizElement);
</script>

{%endblock%}

```

Home.html:

```

{%extends"base.html"%}
{%blocktitle%}Home{%endblock%}
{%blockcontent%}
<style>
/*Optionalbackgroundgradient*/ body
{
    background:linear-gradient(to right,#e0eafc,#cfdef3);
}
.home-container{
    display: flex;
    flex-direction:column;
    align-items: center;
    justify-content:center;
    min-height: 75vh;
    text-align:center;
}
.home-container h2 {
    font-size: 2.5rem;
    margin-bottom:15px;
    color: #003366;
}
.home-container p {
    max-width: 800px;
    font-size: 1.2rem;
    line-height: 1.6;
    margin-bottom:30px;
    color: #333;
}
.get-started-btn {
    display:inline-block;
    padding: 12px 24px;
    font-size: 18px;
    color: white;
    background-color:#007BFF;
    border-radius: 8px;
    text-decoration:none;
    transition:background-color0.3s ease;
}

.get-started-btn:hover {
    background-color:#0056b3;
}

```

}

```

    footer{
      text-align: center;
      padding: 20px;
      background:#f1f1f1;
      margin-top: 40px;
      font-size: 14px;
    }
    @media(max-width:600px){
      .home-containerh2{
        font-size: 2rem;
      }
      .home-containerp{
        font-size: 1rem;
      }
    }
  </style>
  <divclass="home-container">
    <h2>WelcometotheHousingMarketVisualizationApp</h2>
    <p>
      Explorehousingpricetrends,uncoverinsights,andunderstandhowfeatureslikelocation, size,
      and year built impact the cost of homes.
      PoweredbyTableauvisualizationsandauser-friendlyFlask interface.
    </p>
    <a href="{{url_for('dashboard')}}" class="get-started-btn">GetStarted</a>
  </div>
  <footer>
    &copy;2025VisualizingHousing MarketTrends|BuiltwithFlask &Tableau
  </footer>
  {%endblock%}

```

Story.html:

```

{%extends"base.html"%}
{%blocktitle%}Story{%endblock%}
{%blockcontent%}
<h2>HouseTrendsStory</h2>
<!--Tableau StoryEmbed-->
<divclass='tableauPlaceholder'id='viz1751213022339'style='position:relative'>
  <noscript>
    <a href='#'>
      <img alt='House
        Story'src='https://public.tableau.com/static/images/Ho/HousetrendsStory/HouseStory/1_rss.p
        ng'style='border: none' />
    </a>
  </noscript>
  <objectclass='tableauViz'style='display:none;'>
    <paramname='host_url'value='https%3A%2F%2Fpublic.tableau.com%2F'>
    <paramname='embed_code_version'value='3'>
    <paramname='site_root'value='"/>
    <paramname='name'value='HousetrendsStory/HouseStory'>

```

```

<paramname='tabs'value='no'/>
<paramname='toolbar'value='yes'/>
<param
  name='static_image'value='https://public.tableau.com/static/images/Ho/HousetrendsStory/House
  Story/1.png'
/>
<paramname='animate_transition'value='yes'/>
<paramname='display_static_image'value='yes'/>
<paramname='display_spinner'value='yes'/>
<paramname='display_overlay'value='yes'/>
<paramname='display_count'value='yes'/>
<paramname='language'value='en-US'/>
<paramname='filter'value='publish=yes'/>
</object>
</div>
<scripttype='text/javascript'>
  vardivElement=document.getElementById('viz1751213022339');
  varvizElement=divElement.getElementsByTagName('object')[0];
  vizElement.style.width = '100%';
  vizElement.style.height=(divElement.offsetWidth*0.75)+'px'; var
  scriptElement = document.createElement('script');
  scriptElement.src =
  'https://public.tableau.com/javascripts/api/viz_v1.js';vizElement.parentNode.insertBefore(scriptElem
  ent, vizElement);
</script>
{%endblock%}

```

app.py

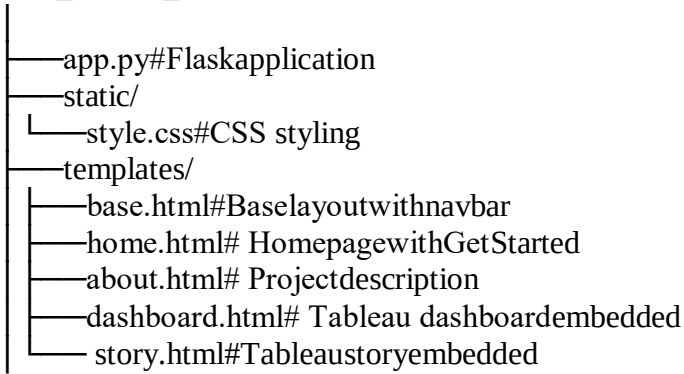
```

fromflaskimportFlask,render_template app
= Flask(name)
@app.route('/')
def home():
    returnrender_template('home.html')
@app.route('/about')
defabout():
    returnrender_template('about.html')
@app.route('/dashboard')
def dashboard():
    returnrender_template('dashboard.html')#TableauJSembedisalreadyinHTML @app.route('/story')
def story():
    return render_template('story.html') #TableauJSembedisalreadyinHTML if
name_____== 'main':
    app.run(debug=True)

```

ProjectStructure

flask_tableau_ui/



DatasetLink

<https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2>

ProjectDemoVideoLink

<https://drive.google.com/file/d/1hknkZkkV-3i7y-I90vYtTKfwzZaEV10V/view?usp=sharing>

GitHubRepositoryLink

<https://github.com/JONNALAGADDASAI SURENDRA/visualizing-house-price-using-Tableau-repo/tree/main/visualizing-house-price-using-Tableau>